

eSim Summer Fellowship 2026

Task 4 — eSim Upgradation Report

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Task	Task 4 — eSim Upgradation
Branch	installers
Repository	https://github.com/aijazvali/eSim/tree/installers

Objective

The aim of this task was to install eSim 2.5 on Ubuntu 25.04, identify the issues that arose during installation, fix as many as possible, and document the entire process in a reproducible manner.

Test Environment

Component	Version / Details
OS	Ubuntu 25.04
Codename	plucky
Kernel	6.14.0-37-generic
Python	3.13.3
Git	2.48.1
GCC	14.2.0
Make	4.4.1
Environment	Virtual machine / Ubuntu desktop

Files Modified

- Ubuntu/install-eSim.sh
- Ubuntu/install-eSim-scripts/install-eSim-24.04.sh

Installation Command

```
bash install-eSim.sh --install
```

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Detailed Issue Report

Issue #1 — Ubuntu 25.04 Not Supported by Top-Level Installer

Status: Fixed

Observation

The main installer script failed immediately on Ubuntu 25.04. It did not select any version-specific installer script, so the installation stopped before reaching the dependency installation steps.

```
Detected Ubuntu Version: Unsupported Ubuntu version: 25.04 ( )
```

Reason

The FULL_VERSION regex expected a three-part version string such as 22.04.4. Ubuntu 25.04 exposed only 25.04, so version extraction failed. The case block in the dispatcher also did not include support for Ubuntu 25.04.

Solution

Changed the FULL_VERSION regex from `\d+\.\d+\.\d+` to `\d+\.\d+(\.\d+)?`, and added a case entry mapping 25.04 to `install-eSim-24.04.sh`.

```
"25.04") SCRIPT="$SCRIPT_DIR/install-eSim-24.04.sh" ;;
```

Result

The installer moved past the unsupported-version failure and started running the version-specific script.

Issue #2 — Invalid apt-get Command for xz-utils

Status: Fixed

Observation

After fixing Ubuntu version detection, the installer failed while preparing volare.

```
Installing volare E: Invalid operation xz-utils
```

Reason

The script used `sudo apt-get xz-utils`, which is not a valid apt-get command. The install subcommand was missing.

Solution

Replaced `sudo apt-get xz-utils` with `sudo apt-get install -y xz-utils`.

```
sudo apt-get install -y xz-utils
```

Result

The installer progressed beyond the volare preparation step.

Issue #3 — Wrong KiCad PPA Selected for Ubuntu 25.04

Status: Fixed

Observation

The installer tried to use the KiCad 6.0 PPA on Ubuntu 25.04.

```
The repository 'https://ppa.launchpadcontent.net/kicad/ kicad-6.0-releases/ubuntu plucky
Release' does not have a Release file.
```

Reason

Inside `installKicad()`, only Ubuntu 24.04 was mapped to KiCad 8.0. Ubuntu 25.04 fell into the fallback branch which used `kicad-6.0-releases`. That PPA does not provide a valid plucky release file.

Solution

Updated the KiCad version-selection logic so that Ubuntu 25.04 no longer falls into the old KiCad 6.0 branch.

Result

The installer stopped selecting the obsolete KiCad 6.0 PPA for Ubuntu 25.04.

Issue #4 — Stale KiCad 6.0 PPA After Failed Run

Status: Fixed

Observation

Even after correcting the KiCad branch logic, `apt update` still failed because the old KiCad 6.0 source was still present on the system.

```
apt update continued to fail due to the old kicad-6.0-releases source in sources.list.d
```

Reason

A previous failed installer run had already added the outdated PPA to `/etc/apt/sources.list.d/`. Changing the installer script did not automatically remove the old source file.

Solution

Removed the stale KiCad 6.0 source file from `/etc/apt/sources.list.d/`. This was a system cleanup step, not a repository code change.

Result

`apt update` no longer failed because of the old KiCad 6.0 repository.

Issue #5 — KiCad 8.0 PPA Dependency Mismatch

Status: Fixed

Observation

The installer failed with unmet dependencies while trying to install KiCad from the KiCad 8.0 PPA.

```
kicad : Depends: libgit2-1.8 (>= 1.8.0) but it is not installable
```

Reason

The KiCad PPA package required `libgit2-1.8`, but Ubuntu 25.04 provides `libgit2-1.9`. The Ubuntu 25.04 repository already contains a compatible distro-native KiCad package.

Solution

Modified `installKicad()` so that Ubuntu 25.04 uses the distro KiCad package instead of the PPA. Also removed the KiCad 8.0 PPA source file.

Result

KiCad 8.0.8 and its required dependencies installed successfully from the Ubuntu 25.04 repository.

Issue #6 — Missing Runtime Asset kicadLibrary.tar.xz

Status: Identified — Not Yet Fixed

Observation

After KiCad installed successfully, the installer failed while extracting the custom KiCad library tarball.

```
tar (child): library/kicadLibrary.tar.xz: Cannot open: No such file or directory tar: Error is not recoverable: exiting now
```

Reason

The installer expects runtime assets such as `library/kicadLibrary.tar.xz`, `images/logo.png`, and `src/frontEnd/Application.py`. The current installers branch does not contain the expected library payload. Git history suggests a branch layout regression.

Solution

No code fix applied yet. Documented as the current blocker.

Result

The issue is identified and documented but still needs to be resolved.

Summary

- Enabled Ubuntu 25.04 support in the top-level installer.
- Fixed malformed apt-get usage for xz-utils.
- Corrected KiCad repository selection logic.
- Resolved stale PPA conflicts from earlier failed runs.
- Switched Ubuntu 25.04 KiCad installation to distro packages.
- Identified missing runtime assets in the installers branch.

Current Status

The installation now proceeds much further on Ubuntu 25.04 than it did initially. Multiple compatibility issues have been fixed, including the Ubuntu version dispatcher, the xz-utils installation command, and the KiCad repository selection path. The remaining blocker is the missing custom KiCad library asset (`library/kicadLibrary.tar.xz`) expected by the installer. Until this asset is restored or the installer is updated to handle its absence, the installation cannot complete fully on Ubuntu 25.04.